superconductivity, Meissner effect, Josephson junctions and applications; Elementary ideas about high temperature superconductivity.

Intrinsic and extrinsic semiconductors; *p-n-p* and *n-p-n* transistors; Amplifiers and oscillators; Op-amps; FET, JFET and MOSFET; Digital electronics-Boolean identities, De Morgan's laws, logic gates and truth tables; Simple logic circuits; Thermistors, solar cells; Fundamentals of microprocessors and digital computers.

POLITICAL SCIENCE AND INTERNATIONAL RELATIONS

PAPER–I

Political Theory and Indian Politics :

1. Political Theory : meaning and approaches.
2. Theories of the State : Liberal, Neo-liberal, Marxist, Pluralist, Post-colonial and feminist.
3. Justice : Conceptions of justice with special reference to Rawl's theory of justice and its communitarian critiques.
4. Equality : Social, political and economic; relationship between equality and freedom; Affirmative action.
5. Rights : Meaning and theories; different kinds of rights; concept of Human Rights.
6. Democracy : Classical and contemporary theories; different models of democracy–representative, participatory and deliberative.
7. Concept of power, hegemony, ideology and legitimacy.
8. Political Ideologies : Liberalism, Socialism, Marxism, Fascism, Gandhism and Feminism.
9. Indian Political Thought : *Dharamshastra, Arthashastra* and Buddhist traditions*;* Sir Syed Ahmed Khan, Sri Aurobindo, M.K. Gandhi, B.R. Ambedkar, M.N. Roy .
10. Western Political Thought : Plato, Aristotle, Machiavelli, Hobbes, Locke, John S. Mill, Marx, Gramsci, Hannah Arendt.

Indian Government and Politics :

1. Indian Nationalism:
   (a) Political Strategies of India's Freedom Struggle: Constitutionalism to mass Satyagraha, Non-cooperation, Civil Disobedience; Militant and revolutionary movements, Peasant and workers' movements.
   (b) Perspectives on Indian National Movement: Liberal, Socialist and Marxist; Radical humanist and Dalit.
2. Making of the Indian Constitution: Legacies of the British rule; different social and political perspectives.

3. Salient Features of the Indian Constitution: The Preamble, Fundamental Rights and Duties, Directive Principles; Parliamentary System and Amendment Procedures; Judicial Review and Basic Structure doctrine.
4. (a) Principal Organs of the Union Government: Envisaged role and actual working of the Executive, Legislature and Supreme Court.
   (b) Principal Organs of the State Government: Envisaged role and actual working of the Executive, Legislature and High Courts.
5. Grassroots Democracy : Panchayati Raj and Municipal Government; significance of 73rd and 74th Amendments; Grassroot movements.
6. Statutory Institutions/Commissions : Election Commission, Comptroller and Auditor General, Finance Commission, Union Public Service Commission, National Commission for Scheduled Castes, National Commission for Scheduled Tribes, National Commission for Women; National Human Rights Commission, National Commission for Minorities, National Backward Classes Commission.
7. Federalism : Constitutional provisions; changing nature of centre-state relations; integrationist tendencies and regional aspirations; inter-state disputes.
8. Planning and Economic Development : Nehruvian and Gandhian perspectives; role of planning and public sector; Green Revolution, land reforms and agrarian relations; liberalilzation and economic reforms.
9. Caste, Religion and Ethnicity in Indian Politics.
10. Party System : National and regional political parties, ideological and social bases of parties; patterns of coalition politics; Pressure groups, trends in electoral behaviour; changing socio- economic profile of Legislators.
11. Social Movements : Civil liberties and human rights movements; women's movements; environmentalist movements.

PAPER – II

Comparative Politics and International Relations

Comparative Political Analysis and International Politics :

1. Comparative Politics : Nature and major approaches; political economy and political sociology perspectives; limitations of the comparative method.
2. State in comparative perspective : Characteristics and changing nature of the State in capitalist and socialist economies, and, advanced industrial and developing societies.
3. Politics of Representation and Participation: Political parties, pressure groups and social movements in advanced industrial and developing societies.